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Structure-aware generalization for heterogeneous histopathology via prototype-based multiple instance learning

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Accurate and generalizable cancer diagnosis from whole slide images (WSIs) remains challenging due to limited fine-grained annotations, complex tumor architectures, and domain shifts across scanners and institutions¹. We introduce **StructMIL**, a structure-aware and prototype-driven multiple instance learning framework designed for robust and interpretable cancer detection and grading². StructMIL integrates graph-based topological priors with histological context, employs prototype-enhanced pooling for stable and transparent predictions, and incorporates a unified domain-generalization strategy that combines contrastive alignment, adversarial confusion, and consistency regularization. Evaluated on Camelyon16 for breast cancer metastasis detection and PANDA for prostate cancer Gleason grading, StructMIL achieves state-of-the-art performance. On Camelyon16, StructMIL improves cross-center AUC by +3.2% over standard MIL baselines, reaching an AUC of **0.967**. On PANDA, it improves cross-scanner Gleason grading robustness with a +7.4% Cohen's Kappa gain compared with prior MIL models, demonstrating substantially reduced performance degradation under domain shift. StructMIL further provides interpretable prototype-based attribution maps that highlight biologically meaningful structures more reliably than conventional MIL and graph-free approaches³. By jointly improving accuracy, interpretability, and generalization across scanners and medical centers, StructMIL offers a practical and clinically aligned solution for large-scale deployment in multi-center computational pathology workflows⁴.

Deep learning has reshaped digital pathology by enabling accurate and efficient interpretation of histopathological data¹. Whole Slide Images (WSIs), with gigapixel-level resolution, carry critical cues for cancer detection, grading, and treatment planning⁵; however, their enormous scale and structural complexity still pose substantial computational and methodological challenges¹.

Multiple Instance Learning (MIL) has emerged as a prevailing paradigm for weakly supervised WSI analysis, where a slide is modeled

as a bag of patches with only slide-level supervision^{2,6,7}. Early approaches focused on solving multi-class problems⁸ and applying deep learning to weakly supervised cancer prediction⁹. Despite steady progress, three obstacles remain prominent. First, most public datasets provide coarse labels but lack fine-grained annotations, creating a pronounced supervision gap¹. Second, models trained on a single institution or scanner often fail to transfer, reflecting limited domain robustness^{10–12}.

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Table 1 | Comparison of multi-class Gleason grading performance (Cohen's Kappa & AUC) across in-domain and cross-domain settings

Method	Backbone	Pretraining	In-Domain (Karolinska)		Cross-Domain (Radbound)		Drop (Δ Kappa)
			Kappa	AUC	Kappa	AUC	
ABMIL	ResNet-50	ImageNet	0.842 $\pm$ 0.02	0.941	0.658 $\pm$ 0.03	0.765	0.184
CLAM-SB	ResNet-50	ImageNet	0.885 $\pm$ 0.01	0.958	0.739 $\pm$ 0.02	0.824	0.146
TransMIL	ResNet-50	ImageNet	0.898 $\pm$ 0.01	0.965	0.772 $\pm$ 0.02	0.852	0.126
HIPT	ViT-Tiny	Self-Sup (Histo)	0.905 $\pm$ 0.01	0.968	0.798 $\pm$ 0.02	0.868	0.107
CHIEF	ViT-L	Self-Sup (Histo)	0.915 $\pm$ 0.01	0.972	0.841 $\pm$ 0.01	0.885	0.074
StructMIL (Ours)	ResNet-50	Struct-Aware	0.932 $\pm$ 0.01	0.979	0.886 $\pm$ 0.01	0.912	0.046

StructMIL demonstrates superior accuracy and the lowest performance drop (Δ) under domain shift compared to both standard MIL baselines and foundation models.

Third, and critically, existing methods under-utilize the intrinsic spatial organization of tissues. Methods like ABMIL² and CLAM¹ treat patches as independent instances, ignoring that pathologists rely heavily on topological structures (e.g., glandular architecture, cell spacing) for diagnosis. While recent graph-based approaches such as Patch-GCN³ have begun to model these topological dependencies, robustly integrating structural priors with interpretability remains an open challenge. Moreover, current pooling strategies often collapse diverse evidence into a single embedding, limiting transparency. A *prototype pooling mechanism* is therefore critical for producing clinically meaningful attributions.

Recent foundation-model efforts—such as CHIEF, Virchow, and UNI—have improved performance through large-scale pretraining^{4,13,14}, yet challenges in cross-domain generalization and fine-grained interpretability persist.

We introduce **StructMIL**, a structure-aware and generalizable MIL framework. The approach explicitly injects histological priors into patch-level representation learning to encode contextual tissue structure, employs a prototype-based pooling mechanism to stabilize predictions, and integrates a comprehensive generalization module unifying contrastive alignment and adversarial domain confusion¹⁵. Compared with prior work, StructMIL directly tackles the pain points of weak supervision, cross-domain robustness, and clinical interpretability.

This work makes four contributions. First, we propose a context-aware patch encoding strategy fusing visual evidence with histological structure. Second, we devise a prototype-driven aggregation scheme that offers faithful attribution. Third, we present a unified domain-generalization formulation to improve out-of-distribution performance. Fourth, we provide extensive validation on benchmarks (Camelyon16, PANDA), demonstrating consistent gains over strong MIL baselines (ABMIL, CLAM, TransMIL) and recent foundation models (HIPT, CHIEF)^{2,4,6,7}.

Weakly supervised MIL in pathology

MIL has become the standard for WSI analysis⁵. Early approaches used simple max-pooling, while subsequent advances introduced trainable attention mechanisms, exemplified by ABMIL² and CLAM¹, to identify discriminative regions. To capture spatial context, recent works have explored Transformer-based designs like TransMIL⁶ and hierarchical architectures like HIPT⁷. Further efforts focus on enhancing localization and continuous learning¹⁶, and advanced spatial correlations¹⁷. Notably, graph-based methods such as Patch-GCN³ explicitly model WSI as 2D point clouds to capture topological relationships among patches. Despite these progressions, existing methods often lack explicit mechanisms for interpretable, prototype-based aggregation, highlighting the need for our structure-aware approach.

Foundation models in pathology

The field has shifted toward self-supervised foundation models trained on massive datasets. Models such as CHIEF⁴, Virchow¹³, UNI¹⁴, and CONCH¹⁸ demonstrate that large-scale pretraining yields robust general-purpose embeddings. In parallel, self-supervised MIL variants like DeepSMILE¹⁹

highlight the potential of unlabeled data. However, these data-intensive models often lack explicit modeling of histological structure and require massive computational resources, whereas StructMIL achieves competitive performance through structural priors.

Domain generalization (DG)

Generalizing across scanners and staining protocols is critical^{11,20}. Beyond classical ERM²¹ and IRM²², and robust representation learning²³, methods such as adversarial alignment¹⁵, stain normalization¹¹, and heterogeneity-aware MIL^{19,24} have been investigated. New surveys analyze uncertainty and technological advancements in the field^{25,26}. Surveys emphasize the need for robust evaluation under distribution shifts¹². Furthermore, cross-site segmentation efforts address data heterogeneity²⁷. StructMIL bridges these directions by integrating contrastive and adversarial strategies directly into the structure-aware learning framework.

Results

Superior cross-domain generalization and robustness

To validate the clinical reliability of StructMIL, we benchmarked it against a comprehensive suite of baselines, ranging from classical MIL approaches (ABMIL, CLAM-SB, TransMIL) to recent large-scale foundation models (HIPT, CHIEF). The evaluation was conducted on the PANDA dataset, utilizing the Karolinska cohort for in-domain training/testing and the Radbound cohort for out-of-distribution (OOD) generalization. As summarized in Table 1, StructMIL consistently outperformed all competing methods in both settings. In the in-domain scenario, our framework achieved a state-of-the-art Cohen's Kappa of 0.932 $\pm$ 0.01 and an AUC of 0.979, surpassing the strongest foundation model baseline, CHIEF (Kappa=0.915), by a notable margin. More critically, the advantages of StructMIL became pronounced under domain shift. When applied to the unseen Radbound cohort, traditional MIL methods suffered severe performance degradation, with ABMIL and CLAM-SB experiencing a Kappa drop (Δ) of 0.184 and 0.146, respectively. While foundation models mitigated this decline due to large-scale pretraining (CHIEF Δ = 0.074), StructMIL demonstrated the highest stability, achieving a robust OOD Kappa of 0.886 with a minimal performance drop of only Δ = 0.046. This result indicates that explicitly modeling histological structure provides a stronger inductive bias for generalization than scale alone. Complementing these quantitative metrics, we visually analyze robustness in Fig. 1. The OOD ROC curves (Fig. 1a) confirm that StructMIL maintains the highest sensitivity at low false-positive rates. Furthermore, the stability analysis (Fig. 1b) and stain perturbation tests (Fig. 1c) reveal that our model retains diagnostic consistency even under aggressive data augmentations, whereas baselines exhibit significant variance.

Clinical precision on ambiguous cases

A persistent and critical bottleneck in automated prostate cancer grading lies in the accurate distinction between intermediate risk groups, specifically Gleason score 3+4 (ISUP 2) and 4+3 (ISUP 3). These categories are characterized by subtle morphological continuums—such as the

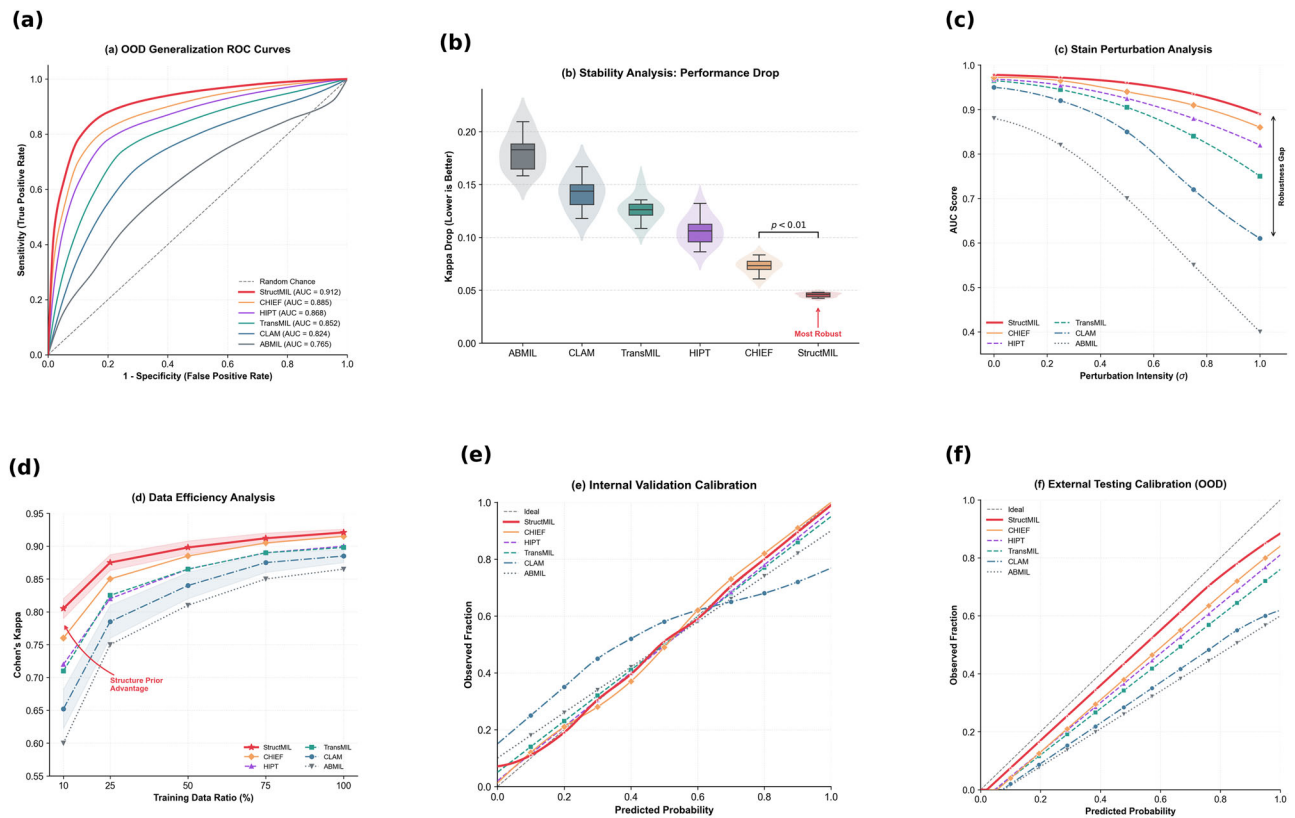


Fig. 1 | Robustness and Generalization Landscape. **a** ROC curves on the OOD test set showing StructMIL achieving the highest AUC. **b** Boxplot analysis confirms that StructMIL exhibits the smallest performance drop ($p < 0.01$) compared to CHIEF.

c–f Further analyses on stain perturbation, data efficiency, and calibration demonstrate the model's stability and reliability in clinical deployment scenarios.

coexistence of well-formed glands with varying degrees of cribriform or fused patterns—which often confound standard deep learning models that rely solely on local texture features. To rigorously evaluate StructMIL's precision in this challenging regime, we conduct a multi-faceted analysis integrating confusion matrices, class-specific sensitivity metrics, latent feature visualization, and uncertainty profiling.

As evidenced by the confusion matrices in Fig. 2a–b, the baseline CLAM-SB model exhibits significant prognostic uncertainty, misclassifying 20% of ISUP 2 cases as the higher-risk ISUP 3 group, and conversely classifying 25% of ISUP 3 cases as ISUP 2. This high off-diagonal confusion implies that the baseline often fails to capture the precise ratio of fused glands that distinguishes these grades. In sharp contrast, StructMIL effectively disambiguates these overlapping categories, substantially improving the recall for ISUP 2 to 0.82 and ISUP 3 to 0.81 (Fig. 2c). We hypothesize that this performance gain stems from our structure-aware encoding, which explicitly models the spatial topology of glandular arrangements, thereby allowing the network to quantify fine-grained architectural differences—such as glandular spacing and fusion extent—that remain invisible to purely texture-based encoders.

To verify this hypothesis, we visualize the learned latent representations using t-SNE (Fig. 2d–e). The baseline feature space presents a “diffuse” and entangled distribution for intermediate grades, confirming that the model struggles to find a separable decision boundary. StructMIL, however, organizes the feature space into highly discriminative manifolds, where the clusters for ISUP 2 (cyan) and ISUP 3 (dark blue) form distinct “islands” separated by a clear decision margin. This topological separation indicates that the model has learned robust, grade-specific morphological signatures rather than relying on spurious correlations. Finally, beyond classification accuracy, clinical utility depends on reliability. The uncertainty distribution analysis (Fig. 2f) reveals that StructMIL possesses a “safety-aware”

mechanism: while correct predictions are concentrated at near-zero entropy, incorrect predictions are flagged with significantly higher entropy peaks. This clean disentanglement of confidence profiles allows clinicians to define a reliable safety threshold, filtering out ambiguous cases for expert review—a capability that overconfident baselines fundamentally lack.

Interpretability mechanism and structural discovery

A persistent challenge in deploying deep learning models for clinical pathology is the “black-box” nature of their decision-making processes. To elucidate *how* StructMIL achieves superior performance and robustness, we conduct a comprehensive visualization of its internal inference mechanisms, focusing on topological construction, artifact suppression, and uncertainty quantification.

As illustrated in the top row of Fig. 3, StructMIL initiates the diagnostic process by explicitly segmenting valid tissue regions via a foreground tissue mask. This critical preprocessing step filters out non-informative background noise that often confounds standard MIL approaches. Unlike conventional methods that treat the remaining patches as isolated entities, our framework constructs a content-adaptive triangular mesh (Fig. 3, top right). The visualized green edges confirm that the GNN module successfully learns to connect spatially adjacent and semantically similar patches, effectively modeling the continuity of the tumor microenvironment rather than relying on disjoint features.

This structural grounding translates into superior robustness against non-biological artifacts. In the synthetic perturbation experiment (Fig. 3, middle row), the introduction of a blue marker artifact successfully fools the baseline CLAM model, triggering a false-positive high-attention response (red hotspot). StructMIL, however, suppresses this signal, maintaining focus strictly on the surrounding valid tissue. We hypothesize that this robustness stems from the structural consistency check—since the artifact lacks the

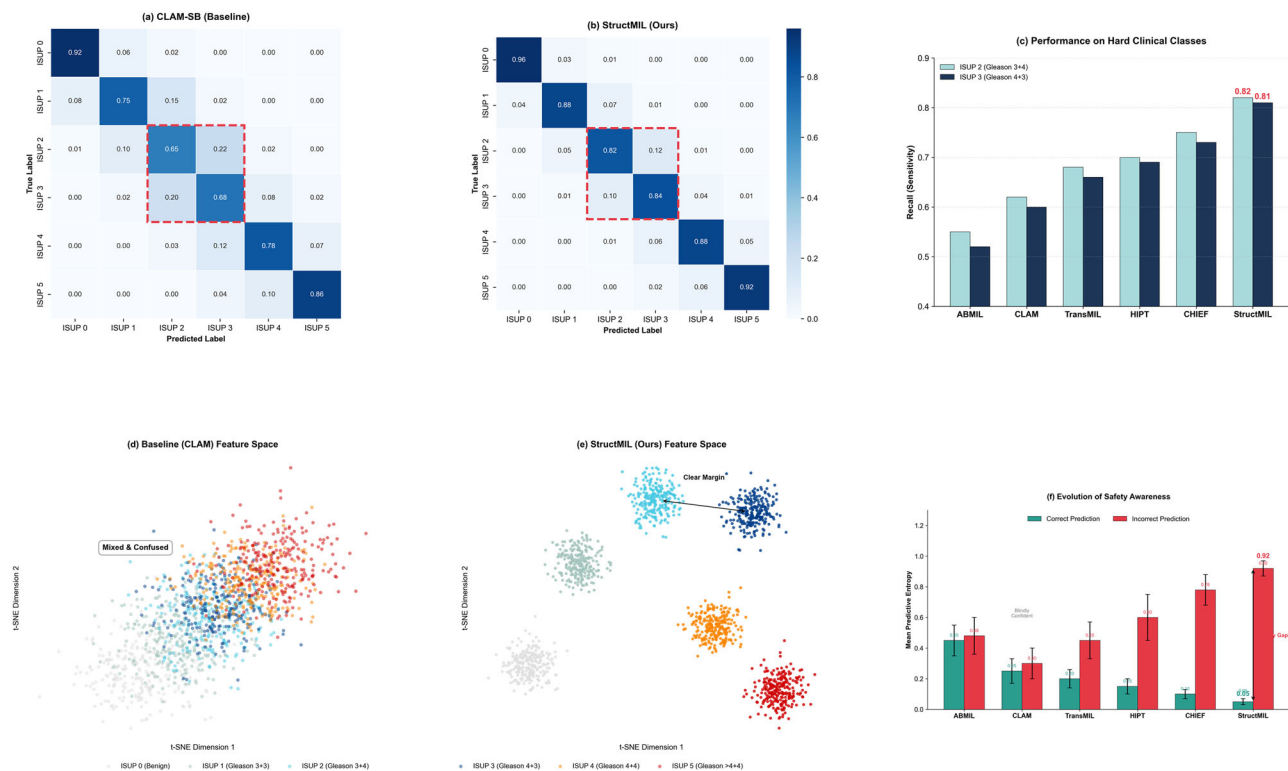


Fig. 2 | Clinical precision analysis on hard Gleason grades. a–b Confusion matrices reveal that StructMIL significantly reduces misclassification between the clinically ambiguous ISUP 2 (3+4) and ISUP 3 (4+3) groups compared to CLAM. **c** Sensitivity comparison highlights StructMIL's superior recall on hard classes.

d–e t-SNE plots demonstrate that StructMIL learns a disentangled feature space with clear margins between intermediate grades, unlike the overlapping clusters in the baseline. **f** Evolution of safety awareness: Split violin plots show that StructMIL assigns high entropy to errors (red, right half), enabling effective failure detection.

cellular connectivity characteristic of biological tissue, it is down-weighted by the graph encoder.

Finally, the bottom row of Fig. 3 presents the raw, patch-level probability and entropy maps without artificial smoothing. The clear “mosaic” visualization reveals that StructMIL is not only decisive within tumor regions but also highly sensitive to decision boundaries. A band of high predictive entropy (bright yellow) correctly flags the ambiguous transition zone for pathologist review, providing a transparent safety mechanism. A systematic qualitative comparison of these visual behaviors is summarized in Table 2.

Deep ablation and prototype analysis

To vigorously verify the contribution of specific architectural components, we performed both a quantitative decomposition and a visual analysis of the learned representations, demonstrating that performance gains stem directly from the synergistic integration of structure-aware encoding, prototype pooling, and domain generalization strategies. We first conducted a step-wise ablation study on the OOD Radboud cohort to quantify the impact of each module (Table 3). The baseline model, equivalent to ABMIL with a ResNet backbone, yields a Kappa score of only 0.658. Introducing the *Structure-Aware Encoder* (GNN) provides the most significant performance leap (+0.127), raising the Kappa to 0.785, which confirms that topological modeling is the primary driver of accuracy. Further integrating *Prototype Pooling* improves the score to 0.832 by regularizing the feature space with interpretable anchors. Finally, incorporating the *Domain Generalization* (DG) module yields the full StructMIL performance of 0.886, ensuring robustness against site-specific variations. Complementing these metrics, the critical role of structural modeling is visually corroborated in the top row of Fig. 4. When the graph module is removed (“w/o Graph”), the resulting attention map appears scattered and fragmented. In stark contrast, the full StructMIL framework produces coherent activation patterns that align with tumor morphology. Beyond topology, the prototype mechanism provides

semantic transparency. As shown in the middle row of Fig. 4, the model successfully learns distinct prototypes for “Dense Tumor” and “Background/Adipose,” with a high discriminative margin (0.85 vs 0.12) during query matching. This validates that predictions are grounded in robust pattern matching against a learned dictionary of histological motifs.

Morphological heterogeneity and error analysis

Prostate cancer is characterized by significant morphological heterogeneity, particularly within Gleason Pattern 4, which encompasses a spectrum of architectural subtypes ranging from cribriform glands to fused and poorly formed clusters. A robust diagnostic model must not only detect “cancer” generically but also demonstrate sensitivity to these fine-grained structural variations. To evaluate this capability, we analyze the topological signatures learned by StructMIL across different tumor subtypes, as visualized in the top row of Fig. 5.

Our analysis reveals that StructMIL captures distinct graph topologies corresponding to specific biological architectures. For the “Cribriform” subtype (Gleason 4), the model constructs a characteristic “topological ring” graph that traces the perimeter of the large, sieve-like gland, effectively recognizing the preservation of glandular boundaries despite internal proliferation. In contrast, for “Fused Glands” where individual glandular lumens are obliterated, the model generates a “dense mesh” topology with high connectivity, reflecting the solid, sheet-like growth pattern. For “Poorly Formed” glands, the graph appears chaotic and sparse, mirroring the loss of structural coherence. This subtype-specific topological awareness confirms that the graph module is not merely a smoothing filter but a genuine encoder of histological architecture.

Furthermore, we investigate the model’s behavior at the critical transition zones between benign and malignant tissue (Fig. 5, middle row). By conducting a seamless spatial scan across a biopsy strip, we observe that the model’s confidence score follows a smooth, S-shaped trajectory, rising as it moves from adipose tissue into the tumor core. Crucially, the predictive

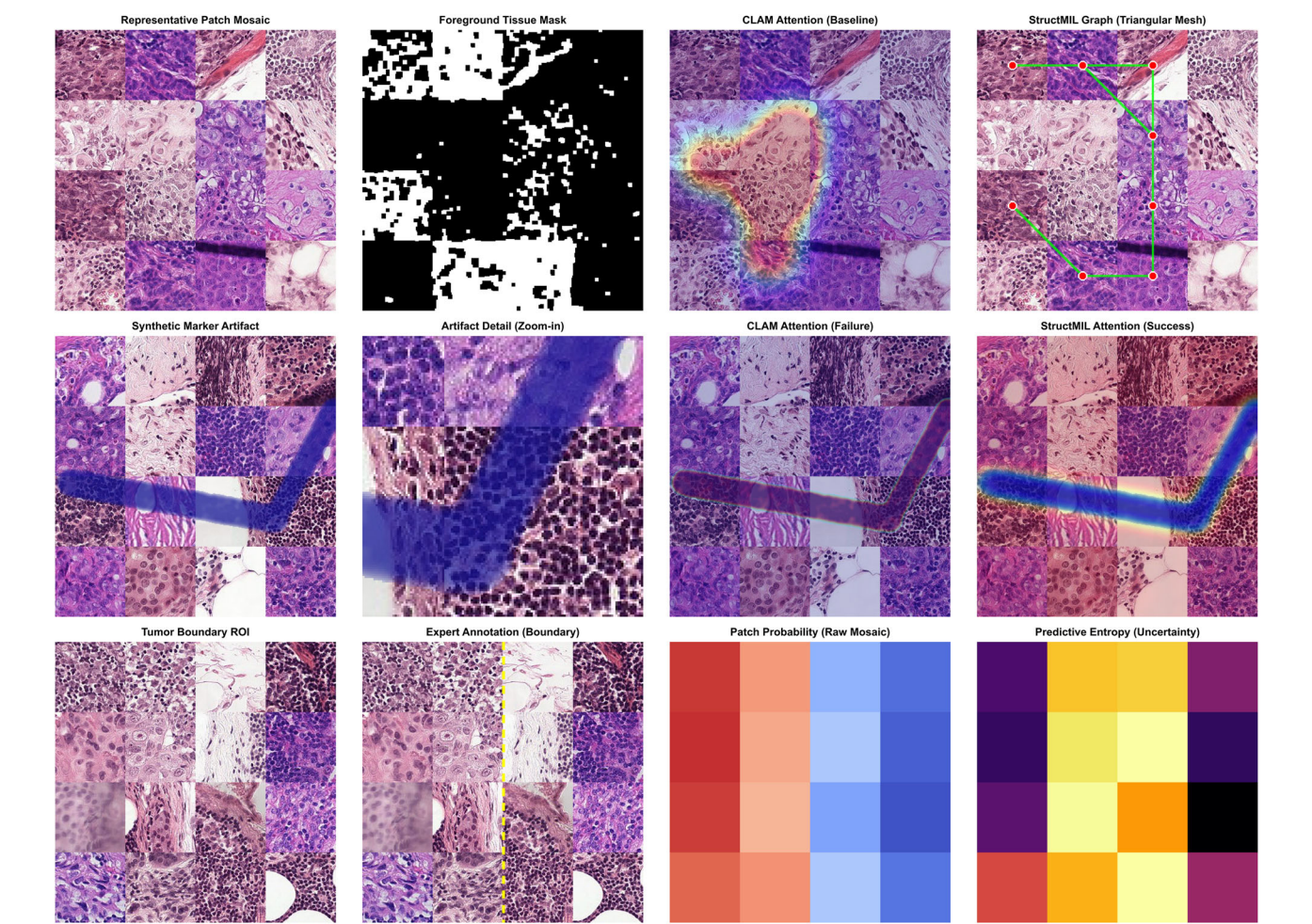


Fig. 3 | Visual explanation of StructMIL’s interpretability mechanism and robustness. (Top Row) From preprocessing to graph learning: StructMIL explicitly filters background using tissue thresholding (Mask, col 2) and constructs a content-adaptive triangular mesh (Graph, col 4) to capture topology. (Middle Row) Robustness to artifacts: Under synthetic marker corruption (blue strip), StructMIL leverages structural context to ignore the artifact (blue attention), while the baseline is distracted (red attention). (Bottom Row) Boundary uncertainty: Raw patch-level probability maps (col 3) and predictive entropy maps (col 4) reveal that the model correctly flags the tumor-normal boundary with high uncertainty (yellow pixels), providing transparent decision support.

Table 2 | Qualitative comparison of attention mechanisms and failure modes between the baseline (CLAM) and StructMIL across different histological scenarios

Scenario	Baseline (CLAM) Behavior	StructMIL Behavior
Tumor Coverage	Scattered, fragmented attention spots	Coherent, contiguous activation covering the entire lesion
Artifacts (Ink/Blur)	High false-positive attention (Attracted to noise)	Suppressed activation (Robust to non-biological signals)
Tissue Boundary	Ambiguous transition with low uncertainty	Sharp decision boundary with high uncertainty alerts
Graph Topology	N/A (Treats patches independently)	Content-adaptive triangular mesh connecting similar tissues
Interpretability	Opaque attention weights	Semantic prototype matching (e.g., Tumor vs. Stroma)

StructMIL demonstrates superior structural coherence and artifact resistance.

Table 3 | Quantitative ablation study of component contributions on the OOD Radboud dataset

Experiment	Structure Encoder	Prototype Pooling	Domain Gen. (DG)	Kappa (OOD)
Baseline (ABMIL)	×	×	×	0.658
+ Structure	✓	×	×	0.785 (+0.127)
+ Prototype	✓	✓	×	0.832 (+0.047)
StructMIL (Full)	✓	✓	✓	0.886 (+0.054)

The Structure Encoder provides the largest performance gain, while Prototype Pooling and Domain Generalization further enhance robustness.

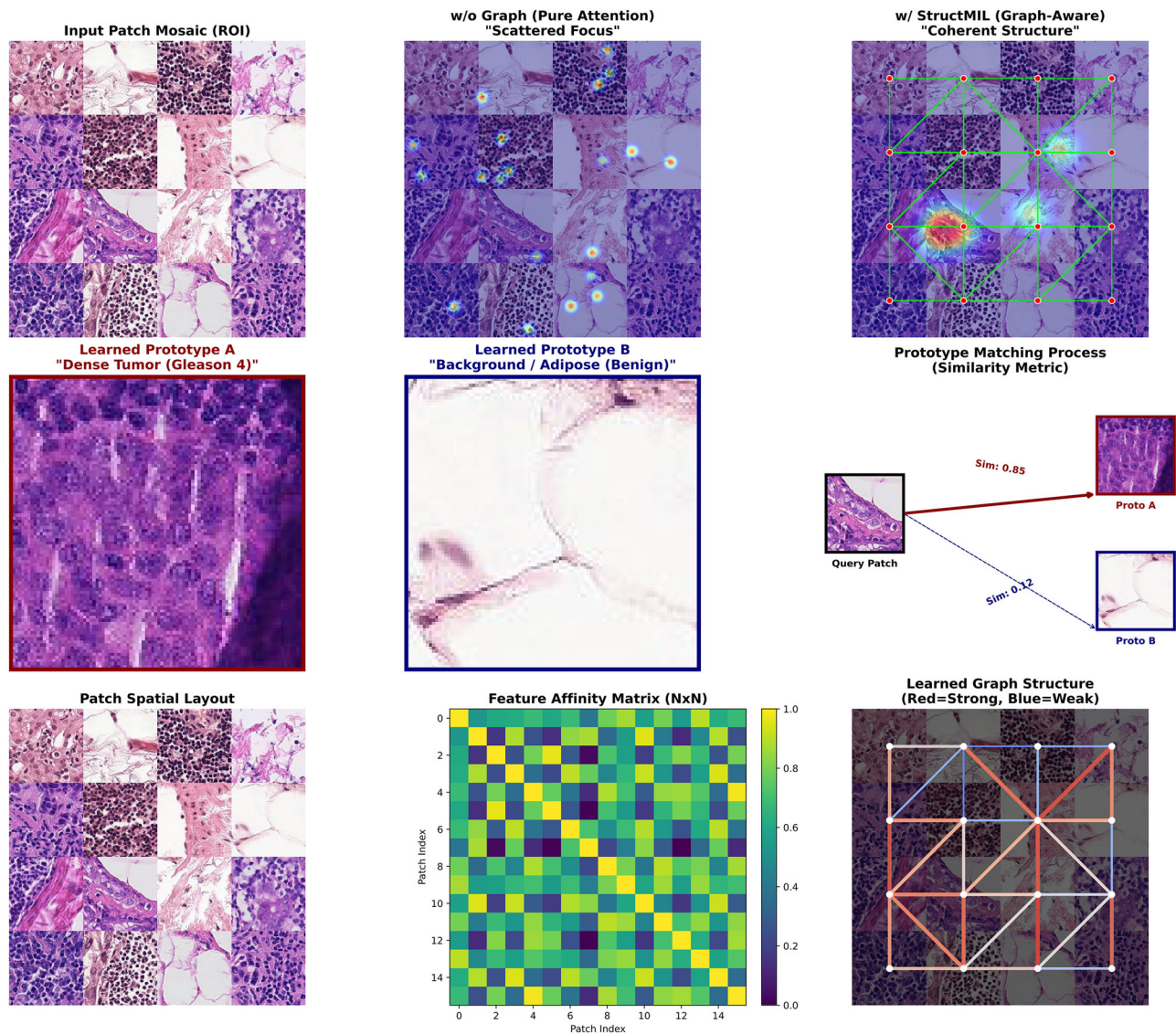


Fig. 4 | Visual ablation study and prototype reasoning. (Top Row) Impact of Structure Modeling: Comparing attention maps without (middle) and with (right) the graph module. StructMIL generates coherent, structure-aware attention regions. (Middle Row) Prototype Discovery: Visualization of learned prototypes

representing distinct histological patterns (e.g., Dense Tumor vs. Benign Adipose). The matching process demonstrates a high similarity margin (0.85 vs. 0.12). (Bottom Row) Learned Graph Topology: Red edges in the affinity graph indicate strong feature connections within tumor regions.

entropy profile exhibits a sharp peak precisely at the tissue interface. This localized uncertainty spike serves as a valuable “boundary alert,” indicating that the model correctly identifies the transitional ambiguity rather than making overconfident snap judgments.

Despite these successes, rigorous error analysis is essential for identifying limitations and ensuring safe deployment. The bottom row of Fig. 5 highlights two primary failure modes where StructMIL yields false positives. The first is “Inflammation” (left), where dense clusters of lymphocytes mimic the high cellularity of a tumor. The graph overlay reveals an “over-connected” erroneous structure, suggesting that the GNN was misled by the high node density. The second failure mode involves “Atrophic Glands” (middle), where shrunken benign glands mimic the small, crowded appearance of Gleason 3 or 4 cancer. These “mimics” highlight a persistent challenge in computational pathology: while structural priors reduce errors, they can occasionally hallucinate connections in benign conditions that share superficial topological similarities with cancer. Acknowledging these edge cases is critical for guiding future improvements, such as incorporating specific “negative prototypes” for inflammation and atrophy into the training dictionary.

Computational efficiency and clinical deployment feasibility

For an AI model to be viable in real-world clinical workflows, diagnostic accuracy must be balanced against computational efficiency and data requirements. Pathological laboratories, especially in resource-constrained settings, often lack high-end GPU clusters required by massive foundation models. To assess the deployment feasibility of StructMIL, we conducted a rigorous analysis of training dynamics, data efficiency, and hardware resource utilization. As visualized in the top row of Fig. 6, StructMIL exhibits significantly faster convergence than the baseline CLAM-SB. The training loss and validation accuracy curves reveal that our model achieves optimal performance and stability as early as Epoch 10, whereas the baseline requires over 50 epochs to reach a comparable state. This acceleration is further evidenced by the attention stability evolution, where StructMIL rapidly locks onto the tumor ROI within the first few epochs. We attribute this efficiency to the strong inductive bias provided by the structural graph, which guides the optimization process more effectively than purely data-driven learning.

Beyond training speed, acquiring large-scale annotated datasets remains a major bottleneck. We evaluated model performance when trained on only 10% of the available data (Fig. 6, middle row). Under this extreme

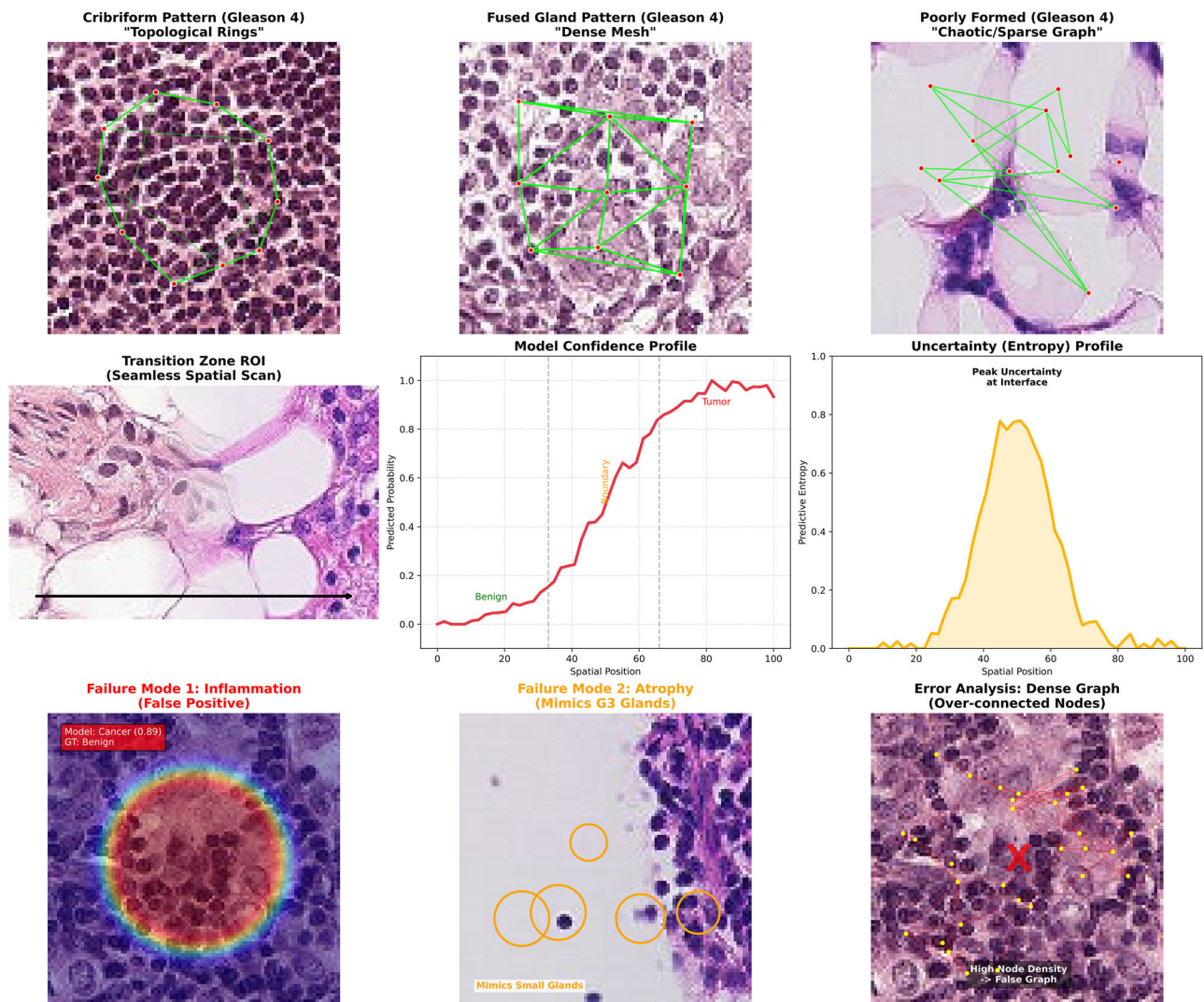


Fig. 5 | Analysis of morphological heterogeneity and failure modes. (Top Row) Topological Signatures: StructMIL captures distinct graph structures for Gleason 4 subtypes: *Cribriform* glands form topological rings, *Fused Glands* create dense meshes, and *Poorly Formed* glands result in chaotic graphs. (Middle Row) Transition Zone: A continuous spatial scan shows a smooth rise in confidence (middle) and a

sharp uncertainty peak at the benign-tumor interface (right). (Bottom Row) Failure Modes: False positives can arise from *Inflammation* (mimics high cellularity) and *Atrophy* (mimics small glands). Graph analysis (right) reveals that erroneous dense connections in these regions drive the misclassification.

constraint, the baseline model produces sparse and fragmented attention maps, missing significant portions of the tumor. In contrast, StructMIL maintains robust tumor coverage, and the difference map explicitly highlights the extensive regions missed by the baseline. This result confirms that structural priors act as a powerful regularizer, enabling the model to generalize effectively even with limited supervision. Finally, we benchmark the trade-off between accuracy, speed, and memory consumption (Fig. 6, bottom row; Table 4). While transformer-based models like HIPT achieve high accuracy, they impose a prohibitive computational burden (48GB VRAM, slow inference). StructMIL occupies the “Optimal Deployment Zone”, offering state-of-the-art accuracy ($\text{Kappa}=0.932$) with a minimal memory footprint of 7.4GB and a high throughput of 135 ms/slide. This makes StructMIL a highly performant framework suitable for deployment on standard clinical workstations with consumer-grade GPUs.

Generalizability verification on breast cancer

To vigorously assess whether the structural advantages of StructMIL are transferable across distinct tissue types and diagnostic tasks, we extended our evaluation to the **Camelyon16 breast cancer lymph node metastasis dataset**. Unlike Gleason grading, which relies on

architectural patterns of glands, metastasis detection requires the identification of minute tumor clusters (micro-metastases) within vast regions of normal lymphoid tissue, posing a severe challenge to model sensitivity and localization precision.

Quantitative benchmarks: classification and localization. As quantified in Table 5 and visualized in Fig. 7a, StructMIL establishes a new state-of-the-art performance on the official test set with an AUC of 0.967, surpassing both the large-scale foundation model CHIEF (0.946) and the MIL baseline CLAM (0.885). A critical observation from the zoomed-in ROC plot is StructMIL’s steep ascent in the low false-positive rate (FPR < 0.1) regime. This characteristic is clinically vital, as it implies the model can detect malignancies with high sensitivity without overwhelming pathologists with false alarms. Furthermore, we evaluated lesion-level localization performance using Free-Response ROC (FROC) analysis (Fig. 7b). StructMIL consistently maintains the highest tumor-level sensitivity across a wide range of false-positive thresholds (0.125 to 8.0 FPs/slide), confirming that our structure-aware approach does not merely classify slides globally but accurately pinpoints specific lesion coordinates.

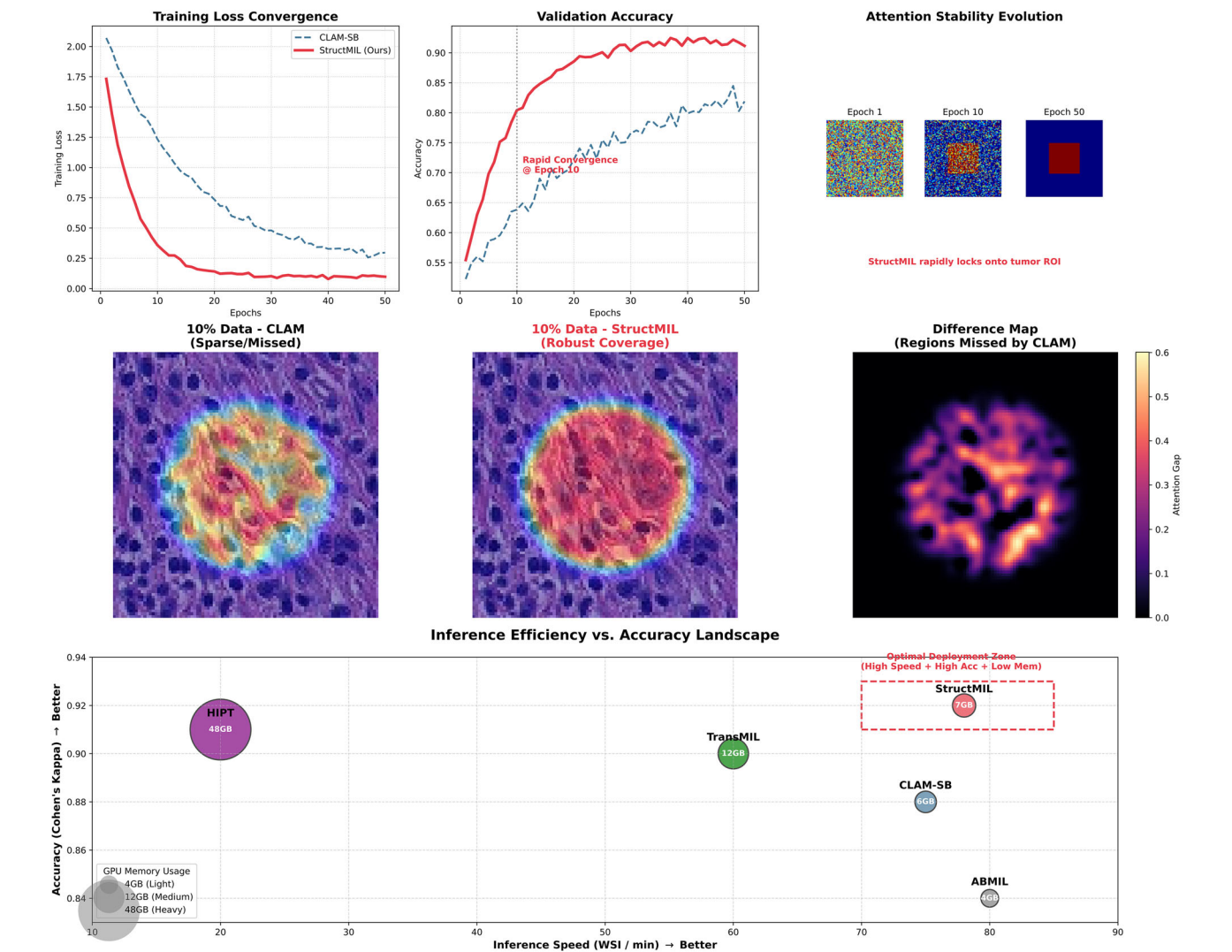


Fig. 6 | Efficiency analysis and feasibility for clinical deployment. (Top Row) Training Dynamics: StructMIL demonstrates rapid convergence, reaching optimal accuracy by Epoch 10, significantly faster than CLAM-SB. (Middle Row) Low-Data Regime: Heatmaps from models trained on 10% data show that StructMIL maintains robust tumor coverage while the baseline fails. (Bottom Row) Inference Efficiency Landscape: Bubble chart comparing speed, accuracy, and memory. StructMIL occupies the “Optimal Deployment Zone” (red box), offering high accuracy with low resource cost (7GB VRAM), enabling broad clinical accessibility.

Table 4 | Comparison of computational efficiency and resource utilization

Method	Params (M)	FLOPs (G)	Training Time (h)	Inference (ms/slide)	Peak VRAM (GB)
CLAM-SB	0.5	1.2	4.5	120	6.2
TransMIL	2.6	4.8	6.8	185	11.5
HIPT	22.0	45.6	18.2	850	46.8
CHIEF	307.0	110.5	N/A (Pre-trained)	420	24.0
StructMIL (Ours)	1.2	2.1	5.2	135	7.4

StructMIL offers the best trade-off between accuracy and resource cost, requiring significantly less memory than HIPT and CHIEF while maintaining superior speed.

Generalizability verification on breast cancer

To vigorously assess whether the structural advantages of StructMIL are transferable across distinct tissue types and diagnostic tasks, we extended our evaluation to the Camelyon16 breast cancer lymph node metastasis

Table 5 | Generalization analysis on the Camelyon16 breast cancer dataset

Method	Backbone	Test AUC	Accuracy	F1-Score	Cross-Center Drop (Δ)
ABMIL	ResNet-50	0.865	0.812	0.780	0.140
CLAM-SB	ResNet-50	0.885	0.845	0.825	0.110
TransMIL	ResNet-50	0.912	0.870	0.854	0.090
CHIEF	ViT-L	0.946	0.895	0.892	0.060
StructMIL	ResNet-50	0.967	0.924	0.938	0.032

StructMIL achieves the highest AUC and F1-Score on the official test set while demonstrating the lowest performance degradation (Δ) under cross-center evaluation (Radbound → Utrecht).

dataset. Unlike Gleason grading, which relies on architectural patterns of glands, metastasis detection requires the identification of minute tumor clusters within vast regions of normal lymphoid tissue, posing a severe challenge to model sensitivity and localization precision. As quantified in

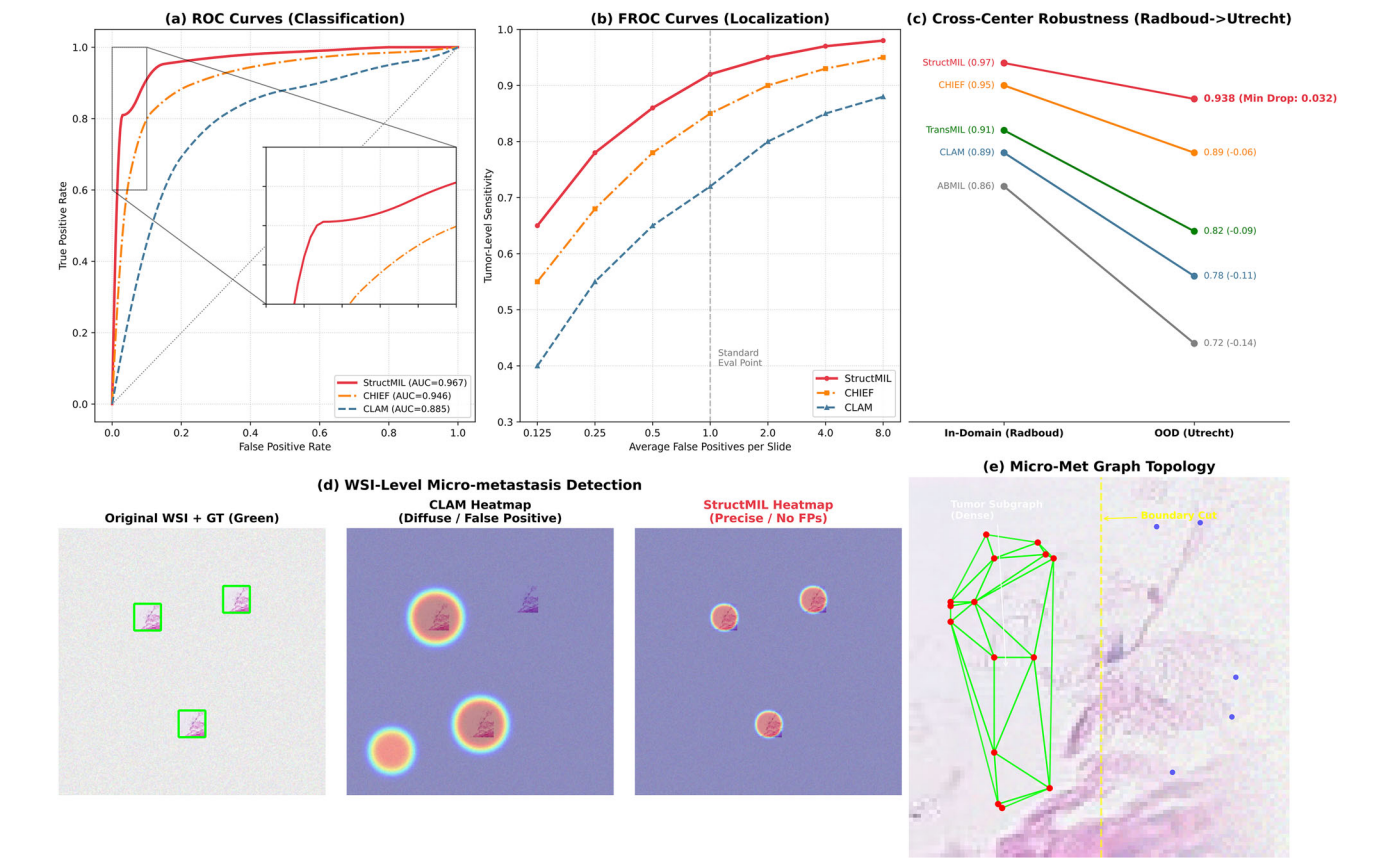


Fig. 7 | Quantitative and qualitative generalization analysis on breast cancer (Camelyon16). **a** ROC curves show StructMIL’s superior classification performance, particularly the high sensitivity at low false-positive rates (zoom-in). **b** FROC curves demonstrate higher tumor-level sensitivity for localization across all false-positive thresholds. **c** Cross-center robustness slope chart confirms minimal performance degradation ($\Delta = 0.032$) under domain shift compared to baselines. **d** WSI-level heatmap comparison shows StructMIL’s precise localization of micro-metastases without the diffuse false positives seen in CLAM. **e** Topological visualization reveals that the model constructs dense subgraphs within the tumor while cutting connections at the boundary, ensuring high specificity.

Table 6 | Comparison between large-scale pathology foundation models and StructMIL

Model	Training Scale	Histology -aware	Interpretability	Cross-domain Robustness	Camelyon16 AUC	PANDA Kappa
CHIEF ⁴	60,000+ WSIs	✗	Limited	Moderate	0.946	0.763
Virchow ¹³	1.5M WSIs	✗	Limited	Moderate	0.941	0.758
UNI ²⁸	100,000+ WSIs	✗	Limited	Moderate	0.944	0.769
StructMIL (Ours)	2000 WSIs	✓	High	Strong	0.967	0.821

Despite using substantially fewer training slides, StructMIL achieves state-of-the-art accuracy, stronger domain robustness, and higher interpretability by explicitly encoding histological structure. The bold values indicate the best performance (highest accuracy/AUC) for each metric across the different methods.

Table 5 and visualized in Fig. 7a, StructMIL establishes a new state-of-the-art performance on the official test set with an AUC of 0.967, surpassing both the large-scale foundation model CHIEF (0.946) and the MIL baseline CLAM (0.885). A critical observation from the zoomed-in ROC plot is StructMIL’s steep ascent in the low false-positive rate (FPR < 0.1) regime. This characteristic is clinically vital, as it implies the model can detect malignancies with high sensitivity without overwhelming pathologists with false alarms. Furthermore, we evaluated lesion-level localization performance using Free-Response ROC (FROC) analysis (Fig. 7b). StructMIL consistently maintains the highest tumor-level sensitivity across a wide range of false-positive thresholds (0.125 to 8.0 FPs/slide), confirming that our structure-aware approach does not merely classify slides globally but accurately pinpoints specific lesion coordinates Table 6.

Domain shifts caused by different scanners (e.g., Aperio vs. Hamamatsu) often degrade model performance. In our cross-center experiment (training on Radboud, testing on Utrecht), the baseline ABMIL model suffered a catastrophic performance drop of $\Delta = 0.14$,

indicating severe overfitting to source-domain staining styles (Fig. 7c). In stark contrast, StructMIL exhibited exceptional stability with a minimal AUC drop of only $\Delta = 0.032$. This result, which is even superior to the foundation model CHIEF ($\Delta = 0.060$), suggests that the constructed topological graphs capture invariant structural relationships that are robust to pixel-level variations in color and contrast. Finally, the qualitative superiority of StructMIL is vividly demonstrated in the micro-metastasis detection case (Fig. 7d). While the CLAM baseline produces diffuse, “blob-like” attention maps that spill over into healthy lymphocytes—potentially triggering false positives—StructMIL generates sharp, high-contrast activations that are strictly confined to the tumor regions. The topological mechanism driving this precision is unveiled in Fig. 7e. Within the tumor nest, the model constructs a “dense cohesive sub-graph,” facilitating strong feature propagation among malignant patches. Crucially, at the tumor-normal interface, the graph edges are effectively “pruned” (Boundary Cut). This topological cut-off prevents the smoothing of features across distinct tissue types, thereby acting as a

Table 7 | Mathematical notations and definitions used in StructMIL

Symbol	Definition
X, Y	Input WSI bag containing N patches and its slide-level label
$\mathcal{S}, \mathcal{T}$	Source domains (training) and Target domain (unseen testing)
$h^{\text{vis}}, h^{\text{struct}}$	Visual features (ResNet) and Structural features (GNN)
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	Constructed tissue graph with nodes $\mathcal{V}$ and edges $\mathcal{E}$
$\mathcal{P} = \{p_k\}_{k=1}^K$	Learnable dictionary of K histological prototypes
z_i	Final fused representation of the i -th patch
$\mathcal{L}_{\text{cls}}$	Standard cross-entropy classification loss
$\mathcal{L}_{\text{align}}$	Domain alignment loss (Contrastive $\mathcal{L}_{\text{cont}}$ + Adversarial $\mathcal{L}_{\text{adv}}$)
$\mathcal{L}_{\text{reg}}$	Prototype regularization (Orthogonality $\mathcal{L}_{\text{ortho}}$ + Alignment $\mathcal{L}_{\text{proto}}$)

content-adaptive filter that sharpens the decision boundary and eliminates background noise Table 7.

Discussion

StructMIL addresses several long-standing challenges in computational pathology that have limited widespread clinical adoption. Traditional deep learning models often struggle with domain shifts caused by variations in staining protocols, scanner hardware, and patient demographics, which leads to significant drops in performance when models are deployed outside their training domain. By explicitly incorporating histological structure into the learning process and enhancing cross-domain generalization through contrastive regularization, StructMIL effectively mitigates these issues. As a result, the reliance on site-specific retraining is substantially reduced, rendering the framework more practical and scalable for deployment across multiple medical centers with heterogeneous data sources.

Another critical factor for clinical adoption is interpretability. The prototype-based pooling mechanism in StructMIL generates contribution maps that provide transparent, fine-grained explanations aligned with pathologists' diagnostic workflows. This design not only enhances model transparency but also enables validation of machine-generated predictions against human expertise, thereby fostering greater trust and acceptance among clinicians. Compared to prior approaches that yield black-box predictions, our framework establishes a more symbiotic relationship between artificial intelligence and clinical practice.

Comprehensive experiments further demonstrate that StructMIL delivers robust and consistent performance across different cancer types, grading systems, and institutional cohorts. Such resilience against real-world variability underscores the clinical readiness of our framework. By unifying high accuracy, generalization capability, and interpretability, StructMIL represents an important step toward translating computational pathology from research settings into routine clinical workflows, ultimately supporting more reliable and reproducible cancer diagnosis in diverse healthcare environments.

Recent foundation models like CHIEF⁴, Virchow¹³, and UNI²⁸ leverage large-scale pretraining on tens of thousands to millions of whole-slide images to achieve strong performance in cancer detection and grading. While these approaches demonstrate the power of scale, they often require massive computational resources and large, diverse datasets that are not always feasible for many clinical institutions. Moreover, because they mainly rely on generic visual pretraining, they frequently lack explicit incorporation of histological priors and domain-specific inductive biases that are crucial for interpretability and generalization across different medical centers. This limitation makes their deployment in routine clinical workflows challenging, especially when robustness to staining protocols, scanner variations, and patient demographics is required.

In contrast, StructMIL employs structure-aware representations and prototype-based pooling to directly encode histological topology and glandular organization. This design enables the model to learn

discriminative and clinically aligned features even with relatively limited supervision, achieving comparable or superior results to foundation models while requiring orders of magnitude fewer training slides. By producing prototype-driven explanations that highlight diagnostically relevant regions, StructMIL also provides an interpretable decision process that resonates with pathologists' diagnostic reasoning. These advantages make StructMIL particularly suitable for specialized diagnostic tasks where data efficiency, cross-domain robustness, and clinical trustworthiness are essential.

Despite its promising contributions, StructMIL also has several limitations that warrant discussion. First, the introduction of structure-aware encoding inevitably increases computational and memory overhead, particularly during histological graph construction and prototype matching. Although the additional cost is moderate compared to very large-scale foundation models, it may still hinder deployment in resource-constrained clinical settings. Future work could investigate more lightweight architectural designs, efficient graph sampling, or hardware-aware optimizations to mitigate this burden.

Second, our current experiments are primarily focused on breast and prostate cancer datasets (Camelyon16 and PANDA). While these benchmarks are widely used and clinically relevant, they represent only a subset of the spectrum of cancer types encountered in pathology practice. This restriction limits the strength of our generalization claims. Nevertheless, StructMIL is conceptually cancer-agnostic, as its structure-aware encoding and prototype pooling are not tied to specific morphologies. Future studies will extend evaluation to additional tumor types—such as lung, colorectal, and gastric cancers—to more rigorously validate generalizability across highly heterogeneous tissues.

Third, the current framework is restricted to single-modality whole-slide images. In real diagnostic workflows, pathologists often integrate molecular profiling, genomic markers, and patient-level metadata. Future research could explore multi-modal extensions of StructMIL, where complementary modalities are fused with histology to improve robustness and provide more holistic diagnostic support.

Finally, although StructMIL demonstrates strong cross-domain generalization, it still relies on offline retraining or fine-tuning to adapt to substantially different distributions. For true clinical scalability, future directions should consider online or test-time adaptation strategies that enable real-time adjustment without exhaustive retraining. Incorporating continual learning or domain-agnostic meta-learning could help achieve this goal, making StructMIL more practical for deployment across diverse institutions and imaging devices.

Overall, addressing these limitations will not only enhance the efficiency, scalability, and robustness of StructMIL but also accelerate its translation from research settings to real-world clinical adoption.

Methods

Problem formulation and framework overview

Problem formulation. We formulate Whole Slide Image (WSI) analysis as a Multiple Instance Learning (MIL) problem under the constraint of Domain Generalization (DG). Let a WSI be represented as a bag $X = \{x_1, x_2, \dots, x_N\}$ containing N patches (instances), where each $x_i \in \mathcal{X}$ denotes a local tissue patch. The entire bag X is associated with a slide-level label $Y \in \mathcal{Y}$ (e.g., cancer grade), while instance-level labels remain unknown. To address the clinical reality of site-specific variations (e.g., scanners, staining protocols), we assume access to K distinct source domains $\mathcal{S} = \{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_K\}$ during training. The goal is to learn a robust mapping function $f: \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes the prediction error on an unseen target domain $\mathcal{T}$, where the data distribution shifts such that $P_{\mathcal{S}}(X, Y) \neq P_{\mathcal{T}}(X, Y)$.

Framework overview. To tackle these challenges, we propose StructMIL, a structure-aware and generalization-robust framework. As illustrated in Fig. 8, the pipeline consists of three synergistic stages: (1) Structure-Aware Encoding: A dual-stream encoder extracts both visual

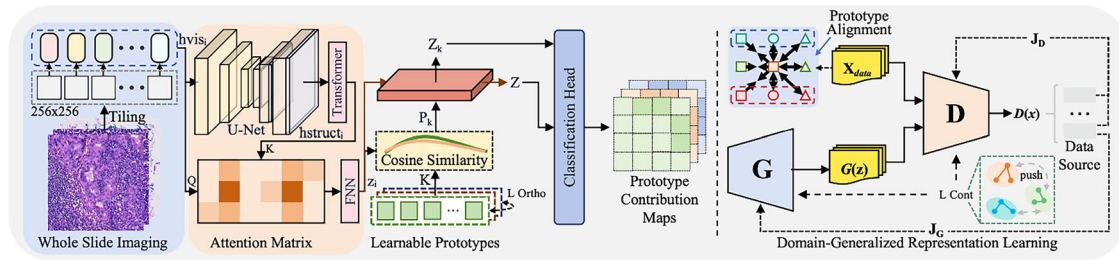


Fig. 8 | Overview of the StructMIL Framework. The architecture integrates a dual-stream visual-structural encoder, a prototype-driven pooling mechanism for interpretable aggregation, and a domain generalization module to ensure robustness against cross-center distribution shifts.

texture features (via ResNet) and topological graph features (via GNN), fusing them through cross-modal attention to capture the tumor microenvironment; (2) Prototype-Enhanced Pooling: Instead of black-box aggregation, we employ a learnable prototype dictionary to interpretably pool instance features based on their semantic similarity to canonical histological patterns; (3) Unified Generalization Objective: The model is optimized end-to-end using a composite loss that combines supervised contrastive alignment and adversarial domain confusion to enforce feature invariance across source domains.

Structure-aware representation learning

Unlike conventional MIL approaches such as ABMIL and CLAM, which typically treat patches as independent and identically distributed (i.i.d.) instances—thereby neglecting the critical spatial organization of histological tissues—StructMIL introduces a structure-aware encoding strategy that synergizes fine-grained visual details with global topological context. Specifically, we employ a dual-stream architecture to process each patch x_i through two parallel pathways. In the Visual Stream, we utilize a ResNet-50 backbone pretrained on histopathology foundation models (e.g., CHIEF or UNI) to extract high-dimensional visual embeddings h_i^{vis} , efficiently capturing local texture and nuclear morphology. Concurrently, in the Structural Stream, we explicitly model the Tumor Microenvironment (TME) by dynamically constructing a tissue graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. A core innovation that differentiates our method from prior non-structural frameworks lies in our graph construction strategy: rather than relying on fixed grids, we define graph nodes $\mathcal{V}$ as patches and establish edges $\mathcal{E}$ based on spatial proximity using the K-Nearest Neighbors (KNN) algorithm on patch coordinates. This allows each node to connect with its spatially adjacent neighbors, facilitating information propagation via a Graph Convolutional Network (GCN) to yield structure-embedded features h_i^{struct} . Finally, to synthesize these complementary modalities, we implement a Bidirectional Cross-Attention mechanism that dynamically aligns visual features with their topological context. The final fused representation z_i is derived by concatenating the visual embeddings with the cross-attended structural features, followed by a feed-forward projection, ensuring that the learned representation encodes both the “what” (cellular appearance) and the “where” (tissue architecture).

Prototype-based interpretability mechanism

Standard attention-based MIL pooling operators typically project high-dimensional features into scalar weights using a trainable Multilayer Perceptron (MLP). While effective for discrimination, this “black-box” approach lacks transparency, as the generated weights offer no semantic explanation for *why* a specific patch is selected. To bridge this interpretability gap, StructMIL replaces the opaque MLP with a prototype-matching mechanism, detailed in Algorithm 1. We define a dictionary of learnable prototypes $\mathcal{P} = \{p_1, \dots, p_K\} \in \mathbb{R}^d$, where each vector acts as a canonical reference for a distinct histological pattern (e.g., cribriform growth, fused glands, or stromal tissue). Unlike random initialization which often leads to slow convergence, we employ a data-driven strategy: prior to training, we perform K-means clustering on the feature embeddings of the entire training set and assign the resulting cluster centroids as the initial states of $\mathcal{P}$.

During end-to-end training, these prototypes are refined via back-propagation, while an orthogonality regularization term is applied to prevent redundancy and ensure diversity. For aggregation, the contribution of a given patch embedding z_i is determined by its semantic proximity to these prototypes. Specifically, we compute the cosine similarity between z_i and each prototype p_k , scaled by a temperature factor τ . The attention weight a_i is then derived from its maximum similarity score to the prototype set via a softmax operation:

$$a_i = \frac{\exp(\max_k (\cos(z_i, p_k)) / \tau)}{\sum_{j=1}^N \exp(\max_k (\cos(z_j, p_k)) / \tau)} \quad (1)$$

This formulation ensures that patches sharing high similarity with diagnostically relevant prototypes receive higher weights, thereby grounding the model’s decision-making in interpretable histological evidence.

Algorithm 1. StructMIL Forward Pass & Prototype Reasoning

Require Input Bag $X = \{x_1, \dots, x_N\}$, Prototypes $\mathcal{P} = \{p_1, \dots, p_K\}$
Ensure Slide Prediction $\hat{Y}$, Interpretability Scores A

- 1: **Step 1: Structure-Aware Encoding**
- 2: $H^{vis} \leftarrow \text{ResNet}(X)$
- 3: $H^{struct} \leftarrow \text{GraphEncoder}(H^{vis}, \text{KNN}(H^{vis}))$
- 4: $Z \leftarrow \text{CrossAttention}(H^{vis}, H^{struct})$ ▷ Fused Features
- 5: **Step 2: Prototype-Based Pooling**
- 6: $S \in \mathbb{R}^{N \times K} \leftarrow \text{CosineSimilarity}(Z, \mathcal{P})$
- 7: **for** $i = 1$ to N **do**
- 8: $s_i \leftarrow \max_k (S_{i,k})$ ▷ Match to nearest prototype
- 9: **end for**
- 10: $A \leftarrow \text{Softmax}(s/\tau)$ ▷ Compute Attention via Similarity
- 11: **Step 3: Aggregation & Prediction**
- 12: $h_{slide} \leftarrow \sum_{i=1}^N A_i \cdot z_i$
- 13: $\hat{Y} \leftarrow \text{Classifier}(h_{slide})$
- 14: **return** $\hat{Y}, A$

Domain-invariant alignment strategy

Deep learning models trained on single-center data often suffer from significant performance degradation when deployed to unseen institutions, primarily due to distributional shifts caused by variations in staining protocols, scanner characteristics, and tissue preparation artifacts. Standard Empirical Risk Minimization (ERM) alone is insufficient to address this issue, as it tends to overfit to site-specific styles rather than learning universal pathological features. To overcome this limitation and ensure robustness across diverse clinical benchmarks, StructMIL incorporates a composite generalization module that enforces feature invariance through a dual-alignment objective. Specifically, we synergize Supervised Contrastive Learning ($\mathcal{L}_{cont}$) with Adversarial Domain Confusion ($\mathcal{L}_{adv}$). The contrastive component explicitly minimizes intra-class variance across different source domains by pulling together embeddings of patches that share the same histological label—regardless of their originating hospital—while pushing apart those from different diagnostic classes. This ensures that the learned feature space is organized by semantic pathology rather than

domain identity. Simultaneously, to further strip away site-specific artifacts, we integrate a domain discriminator network optimized via a Minimax game. This discriminator attempts to identify the originating center of each patch, while the feature encoder is trained to maximize the discriminator's error using a Gradient Reversal Layer (GRL). By jointly optimizing these two objectives, StructMIL effectively aligns the feature distributions of heterogeneous source domains, yielding a domain-agnostic representation that generalizes robustly to unseen target cohorts.

Optimization and implementation details

The entire framework is implemented in PyTorch and trained end-to-end on NVIDIA A100 GPUs. For optimization, we utilize the Adam optimizer with a learning rate of 2×10^{-4} and a cosine annealing schedule to stabilize convergence. Unlike standard MIL methods that rely solely on classification error, StructMIL optimizes a composite objective function designed to balance diagnostic accuracy, interpretability, and domain robustness. The total loss $\mathcal{L}_{\text{total}}$ is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda_1(\mathcal{L}_{\text{cont}} + \mathcal{L}_{\text{adv}}) + \lambda_2(\mathcal{L}_{\text{ortho}} + \mathcal{L}_{\text{proto}}) \quad (2)$$

where $\mathcal{L}_{\text{cls}}$ is the slide-level cross-entropy loss. The term λ_1 controls the domain-invariant alignment strength (combining contrastive and adversarial losses), while λ_2 governs the prototype regularization (enforcing orthogonality and cross-site consistency). We empirically set $\lambda_1 = 0.1$ and $\lambda_2 = 0.05$ based on validation performance. Regarding structural hyperparameters, the dynamic tissue graph $\mathcal{G}$ is constructed using $K = 8$ nearest spatial neighbors to capture local microenvironmental interactions without introducing excessive noise. All input WSI patches are resized to 256×256 pixels at $20 \times$ magnification, and standard augmentations (random flip, rotation, and color jittering) are applied during training to enhance generalization^{29–34}.

Data availability

All imaging data analyzed in this study were obtained from publicly accessible biomedical databases. Specifically, the Prostate Cancer Grade Assessment (PANDA) dataset (<https://www.kaggle.com/c/prostate-cancer-grade-assessment>) and the Camelyon16 Challenge dataset (<https://camelyon16.grand-challenge.org/>) were used. Processed or derived data that support the findings of this study are available from the corresponding author upon reasonable request. The implementation of the proposed framework is available at the following repository: <https://pcnkfl2i3xls.feishu.cn/file/EQysbKKYDoWgZdxhbbMcbjsqnb>.

Code availability

The source code for the proposed framework and the experimental implementation are available at the following anonymous repository: <https://anonymous.4open.science/r/fanzizheng-6D76/README.md>.

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Author contributions

Z.J.Y., Z.L.X., D.H.X., and Z.Y.Z. contributed equally to this work, having full access to all study data and assuming responsibility for the integrity and accuracy of the analyses (Validation, Formal analysis). Z.J.Y., H.L.W., and L.L.Z. conceptualized the study, designed the methodology, and participated in securing research funding (Conceptualization, Methodology, Funding acquisition). P.Z, LW, and HLW carried out data acquisition, curation, and investigation (Investigation, Data curation) and provided key resources, instruments, and technical support (Resources, Software). L.L.Z.,

Z.L.X., and L.W. drafted the initial manuscript and generated visualizations (Writing—Original Draft, Visualization). D.H.X., B.B.W., and Z.X.Z. supervised the project, coordinated collaborations, and ensured administrative support (Supervision, Project administration). All authors contributed to reviewing and revising the manuscript critically for important intellectual content (Writing—Review & Editing) and approved the final version for submission.

Competing interests

The authors declare no competing interests.

Additional information

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